

Challenge Alignment Contract

CURRENT STATE AND FUTURE STRATEGY

- Ratings use the stricter evidence-gated source table
- Current state is separated from future strategy
- Hosted proof is same-day full-grid technical publication, not scientist validation closure
- Candidate methods stay gated until supporting evidence exists
- MVP V2 adds a Himalayan v3 partner-evidence contract, Swiss RAvaFcast reproduction lane, and SAR shadow gates

Evidence level: Artifact/doc proof only

INTERPRETATION

CURRENT STATE

Evidence now

Use only hosted, admin, repo, and artifact proof that exists today.

GATED

Promotion rules

MTS-LSTM, SAR, TreeSHAP refresh, and runout physics require explicit evidence before promotion.

FUTURE STRATEGY

Validation program

Scientist review decides which technical paths become stronger operational claims.

Data Collection And Observation Scarcity

CURRENT STATE AND FUTURE STRATEGY

- Dangerous manual collection: revised rating 3/5
- Sparse AWS networks: revised rating 3/5
- Field reports and news ingest augment the evidence base
- Open-Meteo and snowpack proxies reduce dependence on dense local stations
- Raw public bulletins are useful inputs but not D_tidy-grade training truth
- Future strategy: scientist-reviewed nowcasts, observer evidence, station observations, and event corroboration

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Occurrence Records And Autonomous Evidence

CURRENT STATE AND FUTURE STRATEGY

- Uncertainty in occurrence records: revised rating 3/5
- Current evidence: deduplication, `label_confidence`, and `training_weight`
- News and field reports are staged before they influence training or review
- V2 contract requires `label_source`, `tidy_label_review_basis`, nowcast refs, observer refs, regime, and timing fields
- Future strategy: scientist-owned adjudication rules, regional gold cases, and local holdout release gates

Evidence level: Repo/admin verified plus Artifact/doc proof only

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Rare Events And Feature Discipline

CURRENT STATE AND FUTURE STRATEGY

- Severe class imbalance: revised rating 4/5
- Feature redundancy and overfitting: revised rating 4/5
- Current technical path references rare-event metrics, KMeansSMOTE, and Recursive Feature Elimination
- Public users do not directly see this mitigation, so it remains technical evidence
- Swiss RF4 remains an initial research signal, not paper parity or Himalayan proof
- Future strategy: benchmark slices that expose miss, false-alarm, calibration, and high-danger recall behavior

Evidence level: Repo/admin verified plus Artifact/doc proof only

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Physical Processes And Weak Layers

CURRENT STATE AND FUTURE STRATEGY

- Complex physical processes: revised rating 3/5
- Current state: snowpack proxies, terrain masks, runout seeding, and bulletin framing
- Current active runout proof is analytical Alpha-Beta fallback
- Future strategy: weak-layer plausibility review and validated WhiteboxTools runout path

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FUTURE STRATEGY

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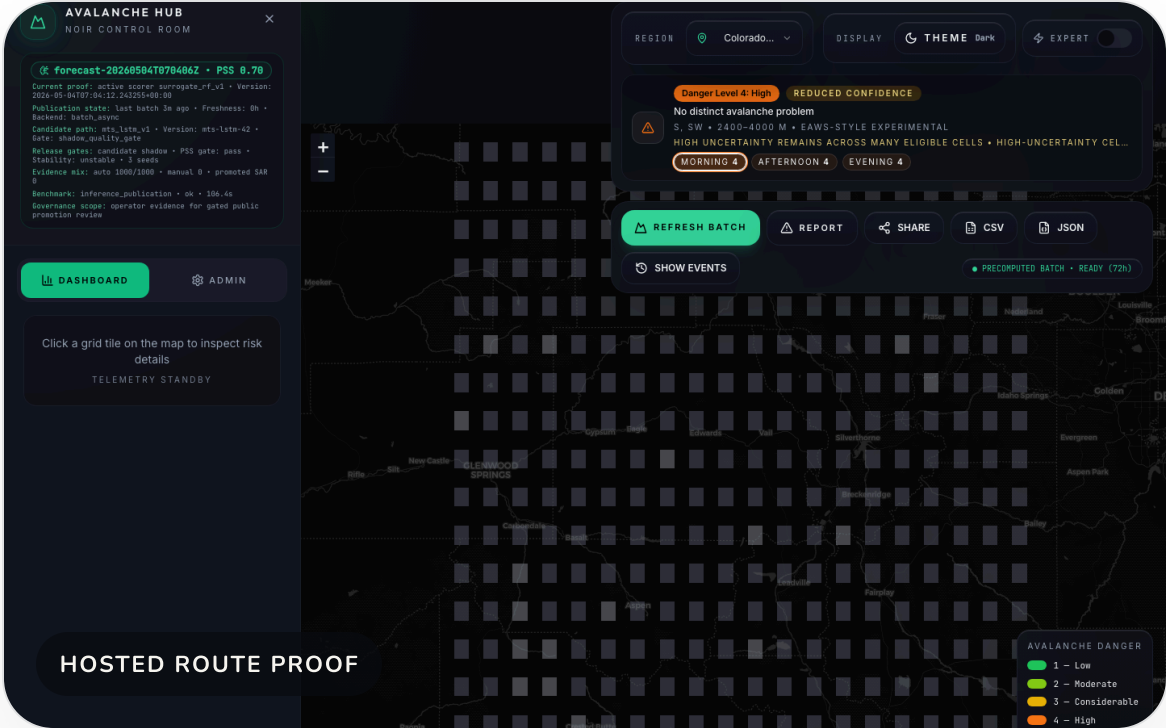
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Spatial And Temporal Hazard Fusion

CURRENT STATE AND FUTURE STRATEGY

- Spatial-temporal disconnect: revised rating 4/5
- Current state: published grid, timeline, daypart bulletin, and optional 3D inspection
- Hosted proof: same-day 20x20 / 72h full-grid technical publication
- RAvaFcast-style GPxyz interpolation requires station latitude, longitude, and elevation_m
- Future strategy: region-specific validation cases, station-density diagnostics, and scientist review of daypart shifts

Evidence level: Hosted production plus Artifact/doc proof only



Weighting, Calibration, And Multiple Optima

CURRENT STATE AND FUTURE STRATEGY

- Subjective parameter weighting: revised rating 3 / 5
- Multiple optima in calibration: revised rating 3 / 5
- Current state: benchmark summaries, PSS/Brier gates, and stability summaries
- Current stability evidence is conservative and should not be framed as closure
- RAvaFcast refined discretization must be learned from training/OOB distributions only, never validation or final holdout labels
- Future strategy: scientist-owned threshold, calibration, and false-alarm tolerance review

Evidence level: Repo/admin verified plus Artifact/doc proof only

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Compute Bottlenecks And Batch Delivery

CURRENT STATE AND FUTURE STRATEGY

- Severe computational bottlenecks: revised rating 4/5
- Current state: manifests, hourly payloads, lazy loading, and Supabase-backed publication metadata
- Modal.com remains off-path candidate compute, not the active public scorer proof
- Future strategy: GitHub Actions or lightweight VPS for repeatable offline publication

Evidence level: Hosted production plus Repo/admin verified

INTERPRETATION

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Disparate Data Integration And Terrain Semantics

- CURRENT STATE AND FUTURE STRATEGY
- Disparate data integration: revised rating 4/5
 - APT masking prevents out-of-scope terrain from looking like ordinary low risk
 - Shared artifacts connect map, bulletin, timeline, and review workflows
 - Future strategy: PostGIS and richer geospatial contracts after validation needs are fixed

Evidence level: Hosted production plus Artifact/doc proof only

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SAR And Remote Sensing Limits

CURRENT STATE AND FUTURE STRATEGY

- Topographic radar shadowing: revised rating 2/5
- Current state: SAR coverage flags and candidate worker paths
- Do not describe SAR as operational or promoted
- Current SAR work is evidence-rich but blocked from production by held-out precision/F1 and fresh-final-holdout gates
- Future strategy: labels, revisit-aware handling, shadow/layover review, and held-out qualification

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Climate Drift And Micro- Climate Variability

CURRENT STATE AND FUTURE STRATEGY

- Climate change concept drift: revised rating 3/5
- Micro-climate variability: revised rating 2/5
- Current state: freshness metadata, regional grids, terrain modifiers, and uncertainty cues
- Future strategy: local validation slices and continuous benchmark refresh

Evidence level: Artifact/doc proof only

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Trust, Explainability, And Reviewability

CURRENT STATE AND FUTURE STRATEGY

- Black-box trust deficit: revised rating 4/5
- Current state: cell inspection, admin provenance, uncertainty messaging, and implemented TreeSHAP path
- Active full-grid run: heuristic_fallback
- Future strategy: refresh active-run TreeSHAP evidence before stronger explanation claims

Evidence level: Repo/admin verified plus Artifact/doc proof only

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Customer Alignment Summary

CURRENT STATE AND FUTURE STRATEGY

- Current state: public route, admin route, full-grid publication proof, workflow controls
- Current state: masked terrain, uncertainty, and current-state bulletin semantics
- Technical evidence: rare-event training discipline and benchmark gates
- MVP V2 evidence: Swiss RAvaFcast reproduction lane, Himalayan v3 partner contract, and one-command partner-package triage
- Future strategy: scientist-led validation, SAR qualification, and candidate-model promotion rules

Evidence level: Hosted production plus Artifact/doc proof only

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Immediate Improvement Areas

CURRENT STATE AND FUTURE STRATEGY

- Full-grid scientist validation over selected regional cases
- SASE/DGRE handoff packet with `partner_handoff_readme.md`, `partner_field_dictionary.md`, checksum guide, source manifest template, and ten blank v3 CSV templates
- One-command triage via `run_himalayan_partner_package_triage`
- TreeSHAP refresh for active full-grid artifacts
- WhiteboxTools runout qualification beyond smoke proof
- SAR held-out evidence and promotion gates
- Field-report and occurrence-record adjudication pack

Evidence level: `Artifact/doc proof only`

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Discussion Close

- CURRENT STATE AND FUTURE STRATEGY
- Approve the current-state MVP as the validation workbench
 - Select priority regions and challenge categories
 - Execute SASE/DGRE partner handoff without sending synthetic rows as evidence
 - Agree benchmark and release-gate ownership
 - Keep customer-facing claims tied to proof buckets

Evidence level: Artifact/doc proof only

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